

EyeOnAI Observability Report

6/13/2026, 9:51:45 PM · tr_84d233d81372 · 3598ms · \$0.0034

QUESTION

Why is the sky blue?

ANSWER

The sky appears blue due to a phenomenon called Rayleigh scattering. When sunlight, which contains all visible wavelengths, passes through the Earth's atmosphere, it encounters gas molecules. These molecules scatter the shorter (blue) wavelengths of light much more than the longer (red) wavelengths. As a result, blue light is dispersed in all directions and reaches our eyes from every part of the sky, giving it a blue appearance. This scattering effect is inversely proportional to the fourth power of the wavelength, which is why blue light (~450nm) scatters roughly 5.5 times more than red light (~700nm).

TRUST SCORE

87/100

Weighted Trust Score

PASS

Groundedness 20% · Hallucination 20% · Relevance 15% · Safety 15%
Clarity 10% · Cost 10% · Citation 5% · Completeness 5%

AGENT EVALUATION — LAYER 2 (INDEPENDENT LLM EVALUATOR)

PASS

Model: gpt-4o-mini-evaluator · 845ms · 1428 tokens

The answer is a solid explanation of why the sky appears blue, relying on the provided evidence and accurately describing the phenomenon of Rayleigh scattering. However, it could benefit from a more concise explanation and a brief mention of the differences in sky color on other planets.

Evaluator Score: **92**/100 · Groundedness 20% · Hallucination 20% · Relevance 15% · Safety 15% · Clarity 10% · Citation 10% · Completeness 10%

METRIC	SCORE	EVALUATOR REASONING
Groundedness	90%	The answer relies heavily on the provided evidence from Encyclopaedia Britannica, NASA Science, and Physics Today, but does not directly quote or reference the exact wording of the evidence.
Hallucination	90%	The answer does not introduce any fabricated facts or invented specifics, but does include a minor inaccuracy in the exact wording of the fourth power relationship.
Relevance	100%	The answer directly addresses the question of why the sky appears blue.
Safety	100%	The answer does not contain any harmful, dangerous, or irresponsible content.
Clarity	90%	The answer is well-structured and understandable to a non-expert, but could benefit from a more concise explanation.
Citation Integrity	80%	The answer makes a claim about the fourth power relationship, but does not directly reference the evidence that supports this claim.
Completeness	90%	The answer fully addresses the question of why the sky appears blue, but could benefit from a brief mention of the differences in sky color on other planets.

Caught by evaluator, missed by rules:

The answer could benefit from a more concise explanation and a brief mention of the differences in sky color on other planets, which would improve its clarity and completeness.

MATHEMATICAL EVALUATION — RULES ENGINE

GROUNDENEDNESS

HALLUCINATION RISK

63%

25%

COST EFFICIENCY

100%

CITATION INTEGRITY

100%

RESPONSE COMPLETENESS

100%

EVIDENCE TRAIL (3 CHUNKS)

CHUNK ID	SOURCE	SIMILARITY	CITED	CONTENT PREVIEW
SCI-001	Encyclopaedia Britannica — Atmospheric Optics	95%	✓ Cited	The sky appears blue due to Rayleigh scattering. Sunlight contains all visible wavelengths. As it passes through the atmosphere, gas molecules scatter shorter (blue) wavelengths far more than longer (...)
SCI-002	NASA Science — Why Is the Sky Blue?	95%	Not cited	Rayleigh scattering intensity is inversely proportional to the fourth power of wavelength. Blue light (~450nm) scatters roughly 5.5x more than red light (~700nm). At sunset, light travels a longer pat...
SCI-003	Physics Today — Atmospheric Light Scattering	85%	✓ Cited	The sky on other planets differs based on atmospheric composition. Mars has a reddish sky due to iron oxide dust. Venus has an orange-yellow sky. The Moon has no atmosphere, so the sky appears black e...

CONFIDENCE ASSESSMENT — LAYER AGREEMENT

RULES ENGINE**44**avg(groundedness,
1-hallucination)**EVALUATOR
AGENT****97**avg(relevance,
safety, clarity)**LOW CONFIDENCE — Layers
disagree**

Δ 53 points · threshold: 25

Evaluator rates quality higher (97%) than the rules engine (44%) — the answer is well-written but may lack direct evidence grounding in the retrieved chunks.

DISAGREEMENT ROOT-CAUSE ANALYSIS

Why the two scoring layers disagree — computed from NLI evidence, sentence verdicts, and evaluator reasoning.

Hallucination

Rules 25%

Eval 100%

Δ 75pt

The evaluator judges the answer as well-written and plausible (scoring 100%), likely because the claims sound authoritative and internally consistent. However, the rules engine found 2 specific claim(s) with no supporting evidence in the retrieved chunks (scoring 25%). This is a classic 'fluent but unverified' pattern — the answer reads as correct but contains claims that cannot be traced to source material.

UNSUPPORTED CLAIMS (2)

When sunlight, which contains all visible wavelengths, passes through the Earth's atmosphere, it encounters gas molecules. entailment 19%

This scattering effect is inversely proportional to the fourth power of the wavelength, which is why blue light (~450nm) scatters roughly 5.5 times more than red light (~700nm). entailment 0% ~450nm · 5.5 times more · ~700nm

PATTERN: Fluency vs Grounding Gap**Overall Quality**

Rules 44%

Eval 97%

Δ 53pt

The evaluator rates overall quality higher (97%) than the rules engine (44%). The evaluator perceives the answer as well-structured and relevant, while the rules engine's lower score reflects gaps in evidence grounding (NLI entailment) and/or unverified numeric claims. The answer may be fluent but lacks full evidential support from the retrieved documents.

PATTERN: Fluency vs Grounding Gap

DEEP VERIFICATION — SENTENCE-BY-SENTENCE ANALYSIS

SENTENCE	VERDICT	ENTAILMENT	CLAIMS
The sky appears blue due to a phenomenon called Rayleigh scattering.	SUPPORTED	100%	—
When sunlight, which contains all visible wavelengths, passes through the Earth's atmosphere, it encounters gas molecules.	UNVERIFIED	19%	—
These molecules scatter the shorter (blue) wavelengths of light much more than the longer (red) wavelengths.	SUPPORTED	99%	—
As a result, blue light is dispersed in all directions and reaches our eyes from every part of the sky, giving it a blue appearance.	SUPPORTED	99%	—
This scattering effect is inversely proportional to the fourth power of the wavelength, which is why blue light (~450nm) scatters roughly 5.5 times more than red light (~700nm).	UNVERIFIED	0%	~450nm ✓ · 5.5 times more ✗ · ~700nm ✓

PIPELINE STEPS

STEP	DURATION	STATUS
CAPTURE	0ms	ok
INTENT_ROUTER	0ms	ok
RETRIEVER	0ms	ok
CHUNKS	0ms	ok

STEP	DURATION	STATUS
LLM_CALL	525ms	ok
TOOL_CALLS	16ms	ok
FINAL_RESPONSE	0ms	ok
QUALITY_CHECK	3056ms	ok